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Inventor(s)

WANG; Huan et al.

SEMICONDUCTOR DEVICES AND MANUFACTURING METHODS OF THE SAME

Abstract

The present disclosure relates to methods, devices, systems, and techniques for manufacturing contact structures in semiconductor devices, e.g., 3D memory devices such as DRAM. An example semiconductor device includes a first semiconductor structure and a second semiconductor structure connected together. The first semiconductor structure includes a first contact structure extending along a first direction and an array of memory cells. The first contact structure includes a first cross section and a second cross section both perpendicular to the first direction. The first cross section is farther away from the second semiconductor structure than the second cross section along the first direction. A size of the first cross section is smaller than a size of the second cross section.

Inventors: WANG; Huan (Wuhan, CN), ZHOU; Chengbao (Wuhan, CN), XIAO; Liang (Wuhan, CN), ZHANG; Mingkang (Wuhan, CN), ZHAO; Yingcheng (Wuhan, CN)

Applicant: Yangtze Memory Technologies Co., Ltd. (Wuhan, CN)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2024/080763, filed on Mar. 8, 2024, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to semiconductor devices and fabrication methods thereof.

BACKGROUND

[0003] Semiconductor industry is driven by the need to produce smaller and faster chips. Makers of memory devices and systems also are pushing to improve scaling techniques. Dynamic random-access memory (DRAM) is a common type of memory device widely used in computer systems. Therefore, advanced techniques for mitigating problems and improving scaling of DRAM devices are desirable.

SUMMARY

[0004] The present disclosure describes methods, devices, systems, and techniques for manufacturing contact structures in semiconductor devices, e.g., 3D memory devices such as DRAM.

[0005] One aspect of the present disclosure features a semiconductor device that includes a first semiconductor structure and a second semiconductor structure connected together. The first semiconductor structure includes a first contact structure extending along a first direction and an array of memory cells. The first contact structure includes a first cross section and a second cross section both perpendicular to the first direction. The first cross section is farther away from the second semiconductor structure than the second cross section along the first direction. A size of the first cross section is smaller than a size of the second cross section.

[0006] In some implementations, the second semiconductor structure includes a peripheral circuit associated with the array of memory cells, and the peripheral circuit is coupled to the first contact structure.

[0007] In some implementations, the size of the first cross section is a maximum size of the first cross section along a second direction perpendicular to the first direction, and the size of the second cross section is a maximum size of the second cross section along the second direction.

[0008] In some implementations, the first contact structure extends through a semiconductor layer of the first semiconductor structure, and the first contact structure is spaced from the array of memory cells along a second direction perpendicular to the first direction.

[0009] In some implementations, the semiconductor device further includes a bonding structure between the first semiconductor structure and the second semiconductor structure. The second semiconductor structure is bonded to a first side of the first semiconductor structure through the bonding structure. The bonding structure includes conductive bonding pads and at least one dielectric material isolating the conductive bonding pads in a second direction perpendicular to the first direction.

[0010] In some implementations, the first semiconductor structure further includes an interconnect

layer between the first contact structure and the bonding structure along the first direction. A first end of the first contact structure is coupled to one of the conductive bonding pads of the bonding structure through the interconnect layer. A second end of the first contact structure is coupled to a first conductive contact structure on a second side of the first semiconductor structure, the second side being opposite to the first side.

[0011] In some implementations, the interconnect layer includes at least an interconnect line extending along the second direction and a vertical interconnect access (VIA) extending along the first direction. The VIA includes a third cross section and a fourth cross section both perpendicular to the first direction. The third cross section is farther away from the second semiconductor structure than the fourth cross section along the first direction. A size of the third cross section is smaller than a size of the fourth cross section.

[0012] In some implementations, at least one of the array of memory cells includes a transistor extending along the first direction and a storage structure. A first end of the storage structure is coupled to the transistor. A second end of the storage structure is coupled to a second conductive contact structure in contact with a trench. The trench extends from the second side of the first semiconductor structure into the first semiconductor structure. Both the first conductive contact structure and the second conductive contact structure include aluminum.

[0013] In some implementations, the second conductive contact structure includes a first portion, a second portion, and a third portion. The first portion and the second portion are connected by the third portion. The first portion and the second portion are perpendicular to the first direction. The first portion is in contact with a bottom of the trench. The third portion is in contact with sidewalls of the trench. The second portion is in contact with a part of the second side that is connected to the sidewalls of the trench.

[0014] In some implementations, a ratio between a first size of the trench in the first direction and a second size of the trench in the second direction is smaller than 0.5.

[0015] In some implementations, the first semiconductor structure further includes a second contact structure extending along the first direction through a semiconductor layer of the first semiconductor structure. The second contact structure is coupled to the peripheral circuit. The first contact structure is configured to transfer a control signal to and from the peripheral circuit. The second contact structure is configured to provide power to the peripheral circuit.

[0016] Another aspect of the present disclosure features a method including forming a first semiconductor structure. The first semiconductor structure includes an array of memory cells. At least one of the array of memory cells includes a transistor extending along a first direction and a storage structure coupled to the transistor. The method further includes forming a contact structure in the first semiconductor structure. The contact structure extends along the first direction. The method further includes bonding the first semiconductor structure to a second semiconductor structure. The second semiconductor structure includes a peripheral circuit associated with the array of memory cells. The contact structure is coupled to the peripheral circuit.

[0017] In some implementations, forming the first semiconductor structure that includes the array of memory cells includes forming a semiconductor body of the transistor in a semiconductor layer of the first semiconductor structure and forming a dielectric region extending through the semiconductor layer along the first direction.

[0018] In some implementations, the first semiconductor structure includes a first side and a second side opposite to the first side. The first side is closer to the semiconductor layer than the second side along the first direction. Forming the contact structure in the first semiconductor structure includes forming a contact hole in the first semiconductor structure by etching the first semiconductor structure from the first side, where the contact hole extends through the dielectric region, and forming the contact structure in the contact hole by depositing a conductive material into the contact hole.

[0019] In some implementations, the method further includes stacking the first semiconductor

structure on a carrier wafer before forming the contact hole, where the second side is in contact with the carrier wafer.

[0020] In some implementations, the method further includes forming an interconnect layer in the first semiconductor structure, where the interconnect layer is coupled to the contact structure and includes at least an interconnect line extending along a second direction perpendicular to the first direction and a VIA extending along the first direction.

[0021] In some implementations, bonding the first semiconductor structure to the second semiconductor structure includes bonding the first semiconductor structure to the second semiconductor structure through a bonding structure. The bonding structure includes conductive bonding pads and at least one dielectric material isolating the conductive bonding pads in a second direction perpendicular to the first direction. The first side is between the second side and the bonding structure.

[0022] In some implementations, the method further includes forming a trench extending from the second side of the first semiconductor structure into the first semiconductor structure and forming a conductive contact structure in the trench by depositing a conductive material in the trench, where the conductive contact structure is coupled to the storage structure.

[0023] A further aspect of the present disclosure features a memory system. The memory system includes a memory device and a memory controller coupled to the memory device and configured to control the memory device. The memory device includes a first semiconductor structure and a second semiconductor structure connected together. The first semiconductor structure includes a first contact structure extending along the first direction and an array of memory cells. The first contact structure includes a first cross section and a second cross section both perpendicular to the first direction. The first cross section is farther away from the second semiconductor structure than the second cross section along the first direction. A size of the first cross section is smaller than a size of the second cross section.

[0024] In some implementations, the size of the first cross section is a maximum size of the first cross section along a second direction perpendicular to the first direction, and the size of the second cross section is a maximum size of the second cross section along the second direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The accompanying drawings, which are incorporated herein and form a part of the present disclosure, illustrate aspects of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person of ordinary skill in the pertinent art to make and use the present disclosure.

[0026] FIG. 1 illustrates a side view of a cross-section of an example three-dimensional (3D) semiconductor device.

[0027] FIGS. 2A-2G show an example fabrication process for manufacturing a semiconductor device.

[0028] FIG. 3 illustrates a flow chart of an example process for manufacturing a semiconductor device.

[0029] FIG. 4 illustrates a block diagram of an example system.

[0030] Like reference numbers and designations in the various drawings indicate like elements. It is also to be understood that the various exemplary implementations shown in the figures are merely illustrative representations and are not necessarily drawn to scale.

DETAILED DESCRIPTION

[0031] Contact structures in a semiconductor device (e.g., a DRAM device) can be configured to provide interconnections between components of the semiconductor device or to couple a

component of the semiconductor device to an outside circuit (e.g., for pad-out purposes). Manufacturing of the contact structures can include forming contact holes in the semiconductor device by an etching process and depositing a conductive material into the contact holes. In some implementations, it can be difficult to form a longer contact structure (e.g., in a vertical direction) and a shorter contact structure (e.g., in the vertical direction) by a same manufacturing process or manufacturing loop. One of the reasons is that when forming a contact hole for the shorter contact structure, the etching process in the manufacturing loop may cause loss of a conductive material such as Tungsten (W) in the semiconductor device. However, forming the longer contact structure and the shorter contact structure using multiple etching masks and multiple etching processes may increase the manufacturing cost and reduce the production yield.

[0032] Implementations of the present disclosure provide techniques for forming a semiconductor device that includes a first semiconductor structure and a second semiconductor structure connected together. Contact structures that extend through the first semiconductor structure and vertical interconnect access (VIA) contacts in an interconnect layer of the first semiconductor structure can be formed in a same manufacturing loop. For example, the contact structures and the VIA contacts can be formed before the first semiconductor structure is bonded to the second semiconductor structure.

[0033] Implementations of the present disclosure can provide one or more of the following technical advantages and/or benefits. Contact structures of different sizes along a vertical direction can be formed by different manufacturing loops, thereby avoiding a Tungsten loss problem. A bottom top metal (BTM) and a bottom top VIA (BTV) coupled to the BTM can be combined into one conductive structure, thereby reducing the manufacturing cost and increasing the production yield. Furthermore, the conductive structure that replaces the BTM and the BTV can be made of aluminum (Al) instead of W, thereby further reducing the manufacturing cost.

[0034] The techniques can be applied to various types of semiconductor devices, volatile memory devices, such as DRAM memory devices, or non-volatile memory (NVM) devices, such as NAND flash memory, NOR flash memory, resistive random-access memory (RRAM), phase-change memory (PCM) such as phase-change random-access memory (PCRAM), spin-transfer torque (STT)-Magnetoresistive random-access memory (MRAM), among others. The techniques can also be applied to charge-trapping based memory devices, e.g., silicon-oxide-nitride-oxide-silicon (SONOS) memory devices, and floating-gate based memory devices. The techniques can be applied to three-dimensional (3D) memory devices. The techniques can be applied to various memory types, such as SLC (single-level cell) devices, MLC (multi-level cell) devices like 2-level cell devices, TLC (triple-level cell) devices, QLC (quad-level cell) devices, or PLC (penta-level cell) devices. Additionally or alternatively, the techniques can be applied to various types of devices and systems, such as secure digital (SD) cards, embedded multimedia cards (eMMC), or solid-state drives (SSDs), embedded systems, among others.

[0035] The details of one or more implementations of the subject matter of this present disclosure are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

[0036] FIG. 1 illustrates a side view of a cross-section of an example 3D semiconductor device **100**. The 3D semiconductor device **100** can be a 3D dynamic random-access memory (DRAM). It is understood that FIG. 1 is for illustrative purposes only and may not necessarily reflect the actual device structure (e.g., interconnections) in practice. The 3D semiconductor device **100** includes a semiconductor structure **102** and a semiconductor structure **104** connected together. In some implementations, the 3D semiconductor device **100** is a bonded chip including the semiconductor structure **104** stacked over the semiconductor structure **102**. The semiconductor structures **102** and **104** can be jointed at a bonding structure **103** therebetween. The semiconductor structure **104** can have two sides **107** and **109**. The side **107** (e.g., the bottom surface) is closer to the semiconductor

structure **102**. The side **109** (e.g., the top surface) is farther away from the semiconductor structure **102** and is opposite to the side **107**.

[0037] It is noted that X, Y, and Z axes (also referred to as X, Y, and Z directions) are included in FIG. **1** to further illustrate the spatial relationship of various components in a semiconductor device. A substrate of the semiconductor device includes two lateral surfaces extending laterally in the X-Y plane: a top surface on the front side of a wafer on which a component of the semiconductor device can be formed, and a bottom surface on the backside opposite to the front side of the wafer. The Z direction is perpendicular to both the X and Y directions. As used herein, whether one component (e.g., a layer or a device) is “on,” “above,” or “below” another component (e.g., a layer or a device) of the semiconductor device is determined relative to the substrate of the semiconductor device in the Z direction (the vertical direction perpendicular to the X-Y plane, e.g., the thickness direction of the substrate) when the substrate is positioned in the lowest plane of the semiconductor device in the Z direction. The same notion for describing the spatial relationships is applied throughout the present disclosure.

[0038] As shown in FIG. **1**, the semiconductor structure **102** can include a substrate **110**, which can include silicon (e.g., single crystalline silicon, c-Si), SiGe, GaAs, Ge, SOI, or any other suitable materials. The semiconductor structure **102** can include peripheral circuits **112** on and/or in the substrate **110**. In some implementations, the peripheral circuits **112** include a plurality of transistors **114** (e.g., planar transistors and/or 3D transistors). Trench isolations (e.g., shallow trench isolations (STIs)) and doped regions (e.g., wells, sources, and drains of transistors **114**) can be formed on or in the substrate **110** as well. In some examples, the peripheral circuits **112** are formed using complementary metal-oxide-semiconductor (CMOS) technology, and the semiconductor structure **102** can be also formed on a semiconductor die that can be referred to as a control die or a CMOS die **102**.

[0039] In some implementations, the semiconductor structure **102** further includes an interconnect layer **116** above the peripheral circuits **112** to transfer electrical signals to and from the peripheral circuits **112**. The interconnect layer **116** can include a plurality of interconnects (also referred to herein as “contacts”), including lateral interconnect lines and vertical interconnect access (VIA) contacts. The interconnect layer **116** can further include one or more interlayer dielectric (ILD) layers in which the interconnect lines and VIA contacts can form. That is, the interconnect layer **116** can include interconnect lines and VIA contacts in multiple ILD layers. In some implementations, peripheral circuits **112** are coupled to one another through the interconnects in the interconnect layer **116**. The interconnects in interconnect layer **116** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

[0040] As shown in FIG. **1**, the bonding structure **103** can include a bonding layer (also referred to as a bonding structure) **118** and a bonding layer (also referred to as a bonding structure) **120** jointed at bonding interface **106** therebetween. The bonding layer **118** can be above the interconnect layer **116** and the peripheral circuits **112**. The bonding layer **118** can include a plurality of bonding contacts **119** and dielectrics electrically isolating the bonding contacts **119**. The bonding contacts **119** can include conductive materials, such as Cu. The remaining area of the bonding layer **118** can be formed with dielectric materials, such as silicon oxide. The bonding contacts **119** and surrounding dielectrics in the bonding layer **118** can be used for hybrid bonding. The bonding layer **120** can be above the bonding layer **118**. The bonding layer **120** can include a plurality of bonding contacts **121** and dielectrics electrically isolating the bonding contacts **121**. The bonding contacts **121** can include conductive materials, such as Cu. The remaining area of the bonding layer **120** can be formed with dielectric materials, such as silicon oxide. The bonding contacts **121** and surrounding dielectrics in the bonding layer **120** can be used for hybrid bonding. The bonding contacts **121** can be in contact with the bonding contacts **119** at the bonding interface **106**. In some

implementations, the bonding layer **120** includes a dielectric layer opposing memory cells (e.g., DRAM cells) **124** with a bit line **123** positioned between the dielectric layer and the memory cells **124**, as shown in FIG. **1**. The dielectric layer can include the bonding interface **106** having the bonding contacts **121**.

[0041] In some implementations, the bonding layer **118** can be considered as a part of the semiconductor structure **102**, and the bonding layer **120** can be considered as a part of the semiconductor structure **104**. The semiconductor structure **104** can be bonded on top of the semiconductor structure **102** in a face-to-face manner at the bonding interface **106**. In some implementations, the bonding interface **106** is disposed between the bonding layers **120** and **118** as a result of hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. In some implementations, the bonding interface **106** is the place at which bonding layers **120** and **118** are met and bonded. In some examples, the bonding interface **106** can be a layer with a certain thickness that includes the top surface of the bonding layer **118** and the bottom surface of the bonding layer **120**.

[0042] In some implementations, the semiconductor structure **104** further includes an interconnect layer **122** above the bonding layer **120** to transfer electrical signals. The interconnect layer **122** can include a plurality of interconnects, such as mid end of line (MEOL) interconnects and back end of line (BEOL) interconnects. The plurality of interconnects in the interconnect layer **122** can include lateral interconnect lines and VIA contacts. In some implementations, the interconnects in interconnect layer **122** also include local interconnects, such as the bit lines **123** and word line contacts (not shown). The interconnect layer **122** can further include one or more ILD layers in which the interconnect lines and VIA contacts can form. The interconnects in the interconnect layer **122** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

[0043] In some implementations, the peripheral circuits **112** include a word line driver/row decoder coupled to the word line contacts in the interconnect layer **122** through the bonding contacts **121** and **119** in the bonding layers **120** and **118** and the interconnect layer **116**. In some implementations, the peripheral circuits **112** include a bit line driver/column decoder coupled to the bit lines **123** and bit line contacts in the interconnect layer **122** through the bonding contacts **121** and **119** in the bonding layers **120** and **118** and the interconnect layer **116**. In some implementations, the bit line **123** is a metal bit line, as opposed to semiconductor bit lines (e.g., doped silicon bit lines). For example, the bit line **123** may include W, Co, Cu, Al, or any other suitable metals having higher conductivities than doped silicon. In some implementations, the bit line contact is an ohmic contact as opposed to a Schottky contact. In some implementations, the bit line **123** is made of a composite conductive material that can be based on a metallic material (e.g., W, Co, Cu, Al) and a semiconductor material (e.g., Si). For example, the composite conductive material can include metal silicide, e.g., such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon.

[0044] In some implementations, the semiconductor structure **104** includes a DRAM device in which memory cells are provided in the form of an array of DRAM cells **124** above the interconnect layer **122** and the bonding layer **120**. That is, the interconnect layer **122** including the bit lines **123** can be disposed between bonding layer **120** and array of DRAM cells **124**. A bit line **123** in the interconnect layer **122** can be coupled to a string of DRAM cells **124**. In some implementations, the semiconductor structure **104** is formed on a semiconductor die and can be referred to as array die **104**.

[0045] In some implementations, a semiconductor device can include multiple array dies (e.g., the

array die **104**) and a CMOS die (e.g., the CMOS die **102**). The multiple array dies and the CMOS die can be stacked and bonded together. The CMOS die can be respectively coupled to each of the multiple array dies, and can respectively drive each of the multiple array dies to operate in the similar manner as the semiconductor device. The semiconductor device can be any suitable device. In some examples, the semiconductor device includes at least a first wafer and a second wafer bonded face to face. The array die can be disposed with other array dies on the first wafer, and the CMOS die can be disposed with other CMOS dies on the second wafer. The first wafer and the second wafer can be bonded together, thus the array dies on the first wafer can be bonded with corresponding CMOS dies on the second wafer. In some examples, the semiconductor device is a chip with at least the array die and the CMOS die bonded together. In an example, the chip is diced from wafers that are bonded together. In another example, the semiconductor device is a semiconductor package that includes one or more semiconductor chips assembled on a package substrate.

[0046] Each DRAM cell **124** can include a vertical transistor **126** and a capacitor **128** coupled to the vertical transistor **126**. DRAM cell **124** can be a 1T1C cell consisting of one transistor and one capacitor. It is understood that DRAM cell **124** may be of any suitable configurations, such as 2T1C cell, 3T1C cell, etc. The vertical transistor **126** can be a MOSFET used to switch a respective DRAM cell **124**. In some implementations, the vertical transistor **126** includes a semiconductor body **130** (the active region in which a channel can form) extending vertically (e.g., in the Z direction), and a gate structure **136** in contact with one side of semiconductor body **130**. In a single-gate vertical transistor, the semiconductor body **130** can have a cuboid shape or a cylinder shape, and the gate structure **136** can abut a single side of semiconductor body **130** in a plane view, e.g., as shown in FIG. **1**. In some implementations, the vertical transistor **126** has a structure including two or more gates, e.g., a two-gates structure, a three-gates structure, or a gate all around (GAA) structure. In some implementations, the gate structure **136** includes a gate electrode **134** and a gate dielectric **132** laterally between the gate electrode **134** and the semiconductor body **130** in a bit line direction (e.g., in the X direction). In some implementations, the gate dielectric **132** abuts one side of the semiconductor body **130**, and the gate electrode **134** abuts the gate dielectric **132**.

[0047] As shown in FIG. **1**, in some implementations, the semiconductor body **130** has two ends (the upper end and lower end in FIG. **1**) in the vertical direction (e.g., the Z direction), and at least one end (e.g., the lower end) extends beyond gate dielectric **132** in the vertical direction (e.g., the Z direction) into the ILD layers. In some implementations, one end (e.g., the upper end) of the semiconductor body **130** is flush with the respective end (e.g., the upper end) of the gate dielectric **132**. In some implementations, both ends (the upper end and lower end) of the semiconductor body **130** extend beyond the gate electrode **134**, respectively, in the vertical direction (e.g., the Z direction) into ILD layers. That is, the semiconductor body **130** can have a larger vertical dimension (e.g., the depth) than that of the gate electrode **134** (e.g., in the Z direction), and neither the upper end nor the lower end of semiconductor body **130** is flush with the respective end of the gate electrode **134**. Thus, short circuits between the bit lines **123** and the word lines/gate electrodes **134** or between the word lines/gate electrodes **134** and the capacitors **128** can be avoided. The vertical transistor **126** can further include a source and a drain (both referred to as **138** as their locations may be interchangeable) disposed at the two ends (the upper end and lower end) of the semiconductor body **130**, respectively, in the vertical direction (e.g., the Z direction). In some implementations, one of the source and drain **138** (e.g., at the upper end in FIG. **1**) is coupled to the capacitor **128**, and the other one of source and drain **138** (e.g., at the lower end in FIG. **1**) is coupled to the bit line **123**. That is, the vertical transistor **126** can have a first terminal in the positive Z direction and a second terminal opposite the first terminal in the negative Z direction, as shown in FIG. **1**.

[0048] In some implementations, the semiconductor structure **104** further includes a substrate **148** disposed above the interconnect layer **122**. The semiconductor body **130** can be formed from the

substrate **148** (e.g., by etching or epitaxy) and thus, has the same semiconductor material as the substrate **148**. The semiconductor body **130** can include semiconductor materials, such as single crystalline silicon, polysilicon, amorphous silicon, Ge, any other semiconductor materials, or any combinations thereof. In one example, semiconductor body **130** may include single crystalline silicon. Source and drain **138** can be doped with N⁺ type dopants (e.g., Phosphorus (P) or Arsenic (As)) or P-type dopants (e.g., Boron (B) or Gallium (Ga)) at a desired doping level. In some implementations, a silicide layer, such as a metal silicide layer, is formed between source/drain **138** of the vertical transistor **126** and the bit line **123** as the bit line contact or between source/drain **138** of the vertical transistor **126** and the first electrode of the capacitor **128** as capacitor contact **142** to reduce the contact resistance. In some implementations, gate dielectric **132** includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al.sub.2O.sub.3, HfO.sub.2, Ta.sub.2O.sub.5, ZrO.sub.2, TiO.sub.2, or any combination thereof. In some implementations, gate electrode **134** includes a conductive material including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicide, or any combination thereof. In some implementations, the gate electrode **134** includes multiple conductive layers, such as a W layer over a TiN layer. In one example, the gate structure **136** may be a “gate oxide/gate poly” gate in which the gate dielectric **132** includes silicon oxide and gate electrode **134** includes doped polysilicon. In another example, gate structure **136** may be an HKMG in which gate dielectric **132** includes a high-k dielectric and gate electrode **134** includes a metal.

[0049] As described above, since the gate electrode **134** may be part of a word line or extend in the word line direction (e.g., the Y direction) as a word line, the second semiconductor structure **104** of the 3D semiconductor device **100** can also include a plurality of word lines each extending in the word line direction (e.g., the Y direction). Each word line **134** can be coupled to a row of DRAM cells **124**. That is, the bit line **123** and the word line **134** can extend in two perpendicular lateral directions, and the semiconductor body **130** of the vertical transistor **126** can extend in the vertical direction perpendicular to the two lateral directions in which the bit line **123** and the word line **134** extend. Word lines **134** are in contact with word line contacts (not shown). In some implementations, the word lines **134** include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, the word line **134** includes multiple conductive layers, such as a W layer over a TiN layer, as shown in FIG. 1.

[0050] In some implementations, as shown in FIG. 1, the vertical transistor **126** extends vertically through and contacts the word lines **134**, and the source or drain **138** of vertical transistor **126** at the lower end thereof is in contact with the bit line **123** (or bit line contact if any). Accordingly, the word lines **134** and the bit lines **123** can be disposed in different planes in the vertical direction due to the vertical arrangement of vertical transistor **126**, which simplifies the routing of the word lines **134** and the bit lines **123**. In some implementations, the bit lines **123** are disposed vertically between the bonding layer **120** and the word lines **134**, and the word lines **134** are disposed vertically between the bit lines **123** and the capacitors **128**.

[0051] As shown in FIG. 1, in some implementations, a capacitor **128** includes a first electrode **144** above and coupled to the source or drain **138** of vertical transistor **126**, e.g., the upper end of the semiconductor body **130**, via a capacitor contact **142**. In some implementations, the capacitor contact **142** is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, the capacitor contact **142** may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon. The capacitor **128** can also include a capacitor dielectric above and in contact with the first electrode **144**, and a second electrode above and in contact with the capacitor dielectric. That is, the capacitor **128** can be a vertical capacitor in which the electrodes and capacitor dielectric are stacked vertically (e.g., in the Z direction), and the capacitor dielectric can be sandwiched between the electrodes. In some implementations, each first electrode is coupled to source or drain **138** of a respective vertical

transistor **126** in the same DRAM cell, while all second electrodes are coupled to a common plate **146** coupled to the ground, e.g., a common ground. The capacitor **128** can have a first end in the negative Z direction and a second end opposite the first end in the positive Z direction, as shown in FIG. **1**. In some implementations, the first end of the capacitor **128** is coupled to the first terminal of the vertical transistor **126** via an ohmic contact (e.g., the capacitor contact **142** made of a metal silicide material).

[0052] It is understood that the structure and configuration of a capacitor **128** are not limited to the example in FIG. **1** and may include any suitable structure and configuration, such as a planar capacitor, a stack capacitor, a multi-fins capacitor, a cylinder capacitor, a trench capacitor, or a substrate-plate capacitor. In some implementations, the capacitor dielectric includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al.sub.2O.sub.3, HfO.sub.2, Ta.sub.2O.sub.5, ZrO.sub.2, TiO.sub.2, or any combination thereof. It is understood that in some examples, a capacitor **128** may be a ferroelectric capacitor used in a FRAM cell, and the capacitor dielectric may be replaced by a ferroelectric layer having ferroelectric materials, such as PZT or SBT. In some implementations, the electrodes include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof.

[0053] As shown in FIG. **1**, the second semiconductor structure **104** can further include a contact structure **147** (also referred to as a pad-out contact structure or a conductive contact structure) in contact with the common plate **146** for coupling the capacitors **128** to outside circuits, e.g., for pad-out purposes. In some implementations, as shown in FIG. **1**, the contact structure **147** can be a conductive pad that is in contact with a trench **149**. The trench **149** can extend from the side **109** into the semiconductor structure **104** and expose the common plate **146**. In some implementations, a ratio between a first size of the trench **149** in the vertical direction (e.g., the Z direction) and a second size of the trench **149** in a horizontal direction (e.g., the X direction) is smaller than 0.5. For example, the ratio can be 0.3. The contact structure **147** is in contact with a bottom surface (e.g., part of the common plate **146**) and sidewalls of the trench **149**. The contact structure **147** can also cover a part of the side **109** as shown in FIG. **1**. In some implementations, as shown in FIG. **1**, the contact structure **147** includes a first portion **147a**, a second portion **147b**, and a third portion **147c**. The first portion **147a** and the second portion **147b** are connected by the third portion **147c**. The first portion **147a** and the second portion **147b** extends in the X-Y plane (e.g., perpendicular to the Z direction). The first portion **147a** is in contact with a bottom of the trench **149**. The third portion **147c** is in contact with sidewalls of the trench **149**. The second portion **147b** is in contact with a part of the side **109** that is connected to the sidewalls of the trench **149**. In some implementations (not shown in FIG. **1**), the contact structure **147** can include one or more conductive pads (also referred to as bottom top metals (BTMs)) that are disposed on the side **109** and one or more conductive VIAs (also referred to as bottom top VIAs (BTVs)) that extend along the Z direction and connect the conductive pads and the common plate **146**. In some implementations, the contact structure **147** includes conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof.

[0054] As shown in FIG. **1**, the semiconductor structure **104** can further include one or more contact structures **150** (also referred to as a pad-out contact structure) extending along the Z direction. The contact structures **150** can also be referred to as through silicon contacts (TSCs) since each contact structure **150** can extend from the side **109** into the semiconductor structure **104** and extend through the substrate **148**. In some implementations, the contact structures **150** can be spaced from the array of DRAM cells **124** along a horizontal direction (e.g., the X direction). In some implementations, the semiconductor structure **104** can include a dielectric spacer (e.g., having silicon oxide) between each contact structure **150** and the substrate **148** to electrically separate the contact structure **150** and the substrate **148**. One end of the contact structure **150** can be connected to a conductive pad **152** disposed on the side **109**, and another end of the contact structure **150** can

be coupled to the interconnect layer **122**. As a result, the peripheral circuits **112** can be coupled to the DRAM cells **124** through the interconnect layers **116** and **122** as well as the bonding layers **120** and **118**, and the peripheral circuits **112** and the DRAM cells **124** can be coupled to outside circuits through contact structures **150** and conductive pads **152**. The contact structure **150** and the conductive pad **152** can include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. The contact structure **150** and the conductive pad **152** can include either the same conductive material or different conductive materials. In some implementations, the contact structure **150** includes W, and the conductive pad **152** includes Al. The contact structures **150** can be configured to transfer electrical signals between the 3D semiconductor device **100** and outside circuits, e.g., for pad-out purposes. For example, one contact structure **150** can be configured to transfer a control signal to and from the peripheral circuit **112**, and another one contact structure **150** can be configured to provide power to the peripheral circuit **112**.

[0055] In some implementations, each of the contact structures **150** can have a shape like a cylinder or a truncated cone that tapers along the Z direction. For example, a cross section of a top portion of the contact structure **150** can have a smaller size than another cross section of a bottom portion of the contact structure **150**. As shown in FIG. 1, the contact structure **150** can have a cross section **154** and a cross section **156** both perpendicular to the Z direction. The cross section **154** is farther away from the semiconductor structure **102** than the cross section **156** along the Z direction. In other words, the cross section **154** is closer to the side **109** of the semiconductor structure **104** than the cross section **156** along the Z direction. A size of the cross section **154** is smaller than a size of the cross section **156**. In some implementations, the size of the cross section **154** is a maximum size of the cross section **154** along a horizontal direction (e.g., the X direction), and the size of the cross section **156** is a maximum size of the cross section **156** along the same horizontal direction (e.g., the X direction).

[0056] In some implementations, a VIA in the interconnect layer **122** (e.g., VIA **158**), which can be referred to as VIA contact or VIA contact structure, also has a shape like a cylinder or a truncated cone that tapers along the Z direction. For example, the VIA **158** can have a top cross section and a bottom cross section (not shown) both perpendicular to the Z direction. The top cross section of the VIA **158** is farther away from the semiconductor structure **102** than the bottom cross section of the VIA **158** along the Z direction. A size of the top cross section of the VIA **158** is smaller than a size of the bottom cross section of the VIA **158**. In some implementations, for example, as described later with further details with respect to FIG. 2C, the VIAs in the interconnect layer **122** and the contact structures **150** are formed by the same manufacturing procedure or manufacturing loop.

[0057] FIGS. 2A-2G show an example fabrication process for manufacturing a semiconductor device (e.g., the semiconductor device **100** of FIG. 1).

[0058] As shown in FIG. 2A, a semiconductor structure **200** can be formed. The semiconductor structure **200** can be an example of the semiconductor structure **104** of FIG. 1, except that some pad-out contact structures (e.g., the contact structure **147**, the contact structures **150**, and the conductive pads **152** of FIG. 1) and at least a part of an interconnect layer (e.g., the interconnect layer **122** of FIG. 1) are not formed in the semiconductor structure **200** yet.

[0059] The semiconductor structure **200** can be formed by any suitable semiconductor manufacturing techniques. The semiconductor structure **200** can include a semiconductor layer **201** and an array of DRAM cells **202** (e.g., DRAM cells **124** of FIG. 1). The semiconductor layer **201** can be a substrate (e.g., substrate **148** of FIG. 1). The array of DRAM cells **202** include vertical transistors **204** (e.g., vertical transistor **126** of FIG. 1) and capacitors **208** (e.g., capacitor **128** of FIG. 1). Each DRAM cell **202** can include a vertical transistor **204** and a corresponding capacitor **208** coupled to the vertical transistor **204**. Each vertical transistor **204** can be a MOSFET used to switch a respective DRAM cell **202**. In some implementations, each vertical transistor **204** includes a respective semiconductor body **210** (e.g., semiconductor body **130** of FIG. 1) extending vertically

(e.g., in the Z direction) and a respective gate structure (e.g., gate structure **136** of FIG. **1**, not shown in FIG. **2A**) in contact with one side of the semiconductor body **210**. The semiconductor bodies **210** can be formed from the substrate **201** (e.g., by etching or epitaxy) and thus, have the same semiconductor material as the substrate **201**. For example, the semiconductor bodies **210** can be formed by forming trenches in the substrate **201** that extend along the lateral directions (e.g., the X direction and the Y direction) during an etching process. The gate structures of the vertical transistors **204** can be formed by depositing a gate dielectric layer and a conductive layer on each semiconductor body **210**.

[0060] As shown in FIG. **2A**, the semiconductor body **210** has two ends in the vertical direction (e.g., the Z direction). The vertical transistor **204** can further include a source and a drain (both referred to as **212** as their locations may be interchangeable) disposed at the two ends (an upper end and a lower end) of the semiconductor body **210**, respectively, in the vertical direction (e.g., the Z direction). In some implementations, the source and drain **212** (e.g., source and drain **138** of FIG. **1**) can be formed by doping the semiconductor body **210** of each vertical transistor **204** with a dopant (e.g., a P-type dopant or an N-type dopant). In some implementations, for each vertical transistor **204**, one of the source and the drain **212** (e.g., at the lower end in FIG. **2A**) is coupled to one of the capacitors **208**, and the other one of source and drain **212** (e.g., at the upper end in FIG. **2A**) is coupled to a bit line **214** (e.g., bit line **123** of FIG. **1**) extending along the X direction. The array of DRAM cells **202** can be between the bit line **214** and the capacitors **208** along the Z direction.

[0061] In some implementations, each of the capacitors **208** can be a vertical capacitor including a first capacitor electrode, a second capacitor electrode, and a capacitor dielectric stacked vertically (e.g., the Z direction). The capacitors **208** can be formed by forming holes (e.g., by etching) extending in the Z direction and being adjacent to the vertical transistors **204**. For each hole, the first capacitor electrode (e.g., a first conductive layer) is formed on the inner surface of the hole, and then the capacitor dielectric (e.g., a dielectric layer in contact with the first conductive layer) and the second capacitor electrode (e.g., a second conductive layer in contact with the dielectric layer) are formed. Each capacitor **208** includes the dielectric layer sandwiched between the first capacitor electrode and the second capacitor electrode. In some implementations, each first capacitor electrode is coupled to source or drain **212** of a respective vertical transistor **204** in the same DRAM cell via a respective capacitor contact **216** (e.g., capacitor contact **142** of FIG. **1**), while the second capacitor electrodes are coupled to a common plate **218** (e.g., common plate **146** of FIG. **1**). The semiconductor structure **200** can also include a plurality of word lines (e.g., word lines **134** of FIG. **1**, not shown in FIG. **2A**) each extending in the word line direction (e.g., the Y direction). Each word line can be coupled to a row of DRAM cells **202**.

[0062] The substrate **201** can include one or more isolation structures **220** extending through the substrate **201** along the Z direction. Each isolation structure **220** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Thus, the isolation structures **220** can also be referred to as dielectric regions. In some implementations, the isolation structures **220** can be formed by etching off the substrate **201**, which can be performed in the same etching process for manufacturing the semiconductor bodies **210**, to form holes that extend through the substrate **201** and then filling the dielectric material into the holes. One or more pad-out contact structures each extending through a corresponding isolation structure **220** along the Z direction can be formed later (e.g., as described with respect to FIGS. **2A-2C**). The isolation structure **220** can surround a pad-out contact structure and isolate the pad-out contact structure from the substrate **201**. In some implementations, the dielectric material of the isolation structure **220** (rather than the silicon of the substrate **201**) will be etched off when forming the pad-out contact structures. The etching process of the isolation structure **220** can be faster than that of the substrate **201**, thereby improving the efficiency of the fabrication process.

[0063] As shown in FIG. 2A, the semiconductor structure **200** can be stacked on a carrier wafer **222**. The semiconductor structure **200** can be bonded to the carrier wafer **222** using any suitable bonding techniques. For example, the semiconductor structure **200** can be bonded to the carrier wafer **222** through an adhesive layer (not shown) disposed therebetween. The adhesive layer can include any suitable types of adhesives. In some implementations, the semiconductor structure **200** can be bonded to the carrier wafer **222** using a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives). The carrier wafer **222** can provide physical support and protection to the semiconductor structure **200** during the fabrication process.

[0064] As shown in FIG. 2B, contact holes **224** and VIA holes **226** extending along the Z direction are formed. Contact holes **224** extend through semiconductor structure **200**. The semiconductor structure **200** can include a first side **228** and a second side **230** opposite to the first side **228**. The first side **228** is closer to the substrate **201** than the second side **230** along the Z direction. The second side **230** can be in contact with the carrier wafer **222**. In some implementations, the contact holes **224** can be formed by etching off the dielectric material in the isolation structures **220** of the semiconductor structure **200** from the first side **228**. The contact holes **224** extend through corresponding isolation structures **220** from the first side **228** to the second side **230**. VIA holes **226** also can be formed by etching off the semiconductor structure **200** from the first side **228**. VIA holes **226** can extend from the first side **228** into the semiconductor structure **200**. In some implementations, each contact hole **224** can have a top cross section **232** and a bottom cross section **234** both perpendicular to the Z direction. The top cross section **232** is closer to the first side **228** than the bottom cross section **234** along the Z direction. A size of the top cross section **232** is larger than a size of the bottom cross section **234**. In some implementations, each VIA hole **226** can have a top cross section **236** and a bottom cross section **238** both perpendicular to the Z direction. The top cross section **236** is closer to the first side **228** than the bottom cross section **238** along the Z direction. A size of the top cross section **236** is larger than a size of the bottom cross section **238**.

[0065] As shown in FIG. 2C, contact structures **240** can be formed by depositing a conductive material into the contact holes **224**, and VIAs **242** can be formed by depositing a conductive material into the VIA holes **226**. In some implementations, the VIAs **242** and the contact structures **240** are formed by the same manufacturing procedure or the same manufacturing loop. The conductive materials can include, but are not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. Contact structures **240** can be examples of contact structures **150** of FIG. 1. VIAs **242** can be examples of VIAs **158** of FIG. 1. VIAs **242** can belong to an interconnect layer **244** (e.g., the interconnect layer **122** of FIG. 1).

[0066] As shown in FIG. 2D, a semiconductor structure **256** can be formed from the semiconductor structure **200** by forming other parts of the interconnect layer **244**, such as interconnect lines **246** extending along a horizontal direction (e.g., the X direction) and VIAs **248** extending along the vertical direction (e.g., the Z direction), on top of the semiconductor structure **200**. The interconnect layer **244** can be coupled to the contact structures **240**. VIAs **242** and VIAs **248** can be made of different materials. For example, VIAs **242** can be made of W, while VIAs **248** can be made of Cu. In some implementations, a bonding layer (also referred to as a bonding structure) **250** can be formed on top of the interconnect layer **244**. As shown in FIG. 2D, the top of the interconnect layer **244** can be considered as a side **245** of the semiconductor structure **256**. The side **245** is opposite to the side **230**. The side **245** is closer to the substrate **201** than the side **230** along the Z direction. The bonding layer **250** can include conductive bonding pads **252** and a dielectric material **254** isolating the conductive bonding pads **252** in a horizontal direction (e.g., the X direction). The semiconductor structure **256** can be an example of the semiconductor structure **104** of FIG. 1.

[0067] As shown in FIG. 2E, the carrier wafer **222** can be removed or de-bonded from the semiconductor structure **256**. The semiconductor structure **256** can be flipped and bonded to a

semiconductor structure **258**. For example, the semiconductor structure **256** can be bonded to the semiconductor structure **258** through a bonding structure **266**. The bonding structure **266** is formed by the bonding layer **250** and a bonding layer (also referred to as a bonding structure) **260** jointed at a bonding interface **269**. The bonding layer **260** is between the bonding layer **250** and the semiconductor structure **258**. The bonding layer **260** can include conductive bonding pads **262** and a dielectric material **264** isolating the conductive bonding pads **262** in a horizontal direction (e.g., the X direction). In some implementations, the bonding layer **250** can be bonded to the bonding layer **260** using a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. That is, the conductive bonding pads **252** in the bonding layer **250** are bonded to the conductive bonding pads **262** in the bonding layer **260**, and the dielectric material **254** in the bonding layer **250** is bonded to the dielectric material **264** in the bonding layer **260**. The bonding layer **250** can be an example of the bonding layer **120** of FIG. 1. The bonding layer **260** can be an example of the bonding layer **118** of FIG. 1.

[0068] In some implementations, the semiconductor structure **256** and the semiconductor structure **258** can be fabricated in parallel. The semiconductor structure **258** can be an example of the semiconductor structure **102** of FIG. 1. For example, the semiconductor structure **258** can include a substrate **267** and peripheral circuits **268** on and/or in the substrate **267**. The peripheral circuits **268** can be an example of the peripheral circuits **112** of FIG. 1. In some implementations, the peripheral circuits **268** include a plurality of transistors **270** (e.g., planar transistors and/or 3D transistors). In some implementations, the semiconductor structure **258** further includes an interconnect layer **272** above the peripheral circuits **268** to transfer electrical signals to and from the peripheral circuits **268**. In some implementations, the peripheral circuits **268** can be configured to control the array of DRAM cells **202**. For example, the peripheral circuits **268** can include a word line driver and/or a bit line driver for the array of DRAM cells **202**. The peripheral circuits **268** can be coupled to the DRAM cells **202** through the interconnect layer **272**, the bonding structure **266**, and the interconnect layer **244**.

[0069] As shown in FIG. 2F, a trench **274** can be formed by an etching process. The trench **274** can extend from the side **230** of the semiconductor structure **256** into the semiconductor structure **256**. The trench **274** can expose the common plate **218** that is coupled to the capacitors **208**. The trench **274** can be an example of the trench **149** of FIG. 1.

[0070] As shown in FIG. 2G, a semiconductor device **280** is formed. The semiconductor device **280** includes the semiconductor structure **256**, the bonding structure **266**, and the semiconductor structure **258**. The semiconductor structure **256** includes conductive pads **276** and contact structure **278**. The conductive pads **276** can be formed by depositing a conductive material on top of the side **230**. Each conductive pad **276** can be connected to an end (e.g., an end on the top) of a corresponding contact structure **240**. The contact structure **278** can be formed by depositing a conductive material in the trench **274** and on top of the side **230**. The conductive pads **276** can be an example of the conductive pads **152** of FIG. 1. The contact structure **278** can be an example of the contact structure **147** of FIG. 1. Each of the conductive pads **276** and the contact structure **278** can be made of conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, the conductive pads **276** and the contact structure **278** are made of Al instead of W, thereby reducing the cost of the manufacturing process. While FIG. 2G shows two conductive pads **276** and one contact structure **278**, it is understood that any suitable number of conductive pads and contact structures can be formed. For example, multiple contact structures **278** can be formed and coupled to the common plate **218** thereby enlarging a contact area between the contact structures **278** and the common plate **218** and reducing the resistance caused by the contact structures **278**. It is also understood that while the conductive pads **276** and the contact structure **278** illustrated by FIG. 2G are merely examples, any other suitable conductive pads and contact structure that are respectively coupled to

the contact structures **240** and the common plate **218** can be formed.

[0071] FIG. **3** illustrates a flow chart of an example process **300**. The process **300** can be performed to form a semiconductor device (e.g., a DRAM). The semiconductor device can be, for example, the semiconductor device **100** of FIG. **1** or the semiconductor device **280** of FIG. **2G**. The process **300** can be described in view of FIGS. **2A-2G**. The process **300** can include the fabrication process of forming the semiconductor structures in FIGS. **2A-2G**. It is understood that the operations shown in process **300** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **3**.

[0072] At operation **302**, a first semiconductor structure (e.g., the semiconductor structure **256** of FIGS. **2D-2G**) is formed. The first semiconductor structure can include an array of memory cells (e.g., the array of DRAM cells **202** of FIG. **2A**). At least one of the array of memory cells includes a transistor (e.g., the vertical transistor **204**) extending along a first direction (e.g., the Z direction) and a storage structure (e.g., the capacitor **208** of FIG. **2A**) coupled to the transistor.

[0073] In some implementations, forming the first semiconductor structure includes forming a semiconductor body (e.g., the semiconductor body **210** of FIG. **2A**) of the transistor in a semiconductor layer (e.g., the substrate **201**) of the first semiconductor structure.

[0074] In some implementations, forming the first semiconductor structure further includes forming a dielectric region (e.g., the isolation structure **220** of FIG. **2A**) extending through the semiconductor layer along the first direction.

[0075] At operation **304**, a contact structure (e.g., the contact structure **150** of FIG. **1** or the contact structure **240** of FIG. **2C**) in the first semiconductor structure is formed. The contact structure extends along the first direction. In some implementations, the first semiconductor structure comprises a first side (e.g., the side **245** of FIG. **2D**) and a second side (e.g., the side **230** of FIG. **2D**) opposite to the first side. The first side can be closer to the semiconductor layer than the second side along the first direction.

[0076] In some implementations, forming the contact structure in the first semiconductor structure includes forming a contact hole (e.g., the contact hole **224** of FIG. **2B**) in the first semiconductor structure by etching the first semiconductor structure from the first side. In some implementations, the contact hole can extend through the dielectric region. In some implementations, forming the contact structure in the first semiconductor structure further includes depositing a conductive material into the contact hole.

[0077] In some implementations, the first semiconductor structure can be stacked on a carrier wafer (e.g., the carrier wafer **222** of FIG. **2A**) before the contact hole is formed. The carrier wafer can provide physical support and protection to the first semiconductor structure during the fabrication process. In some implementations, the second side (e.g., the side **230**) of the first semiconductor structure is in contact with the carrier wafer.

[0078] In some implementations, an interconnect layer (e.g., the interconnect layer **244** of FIG. **2D**) can be formed in the first semiconductor structure. The interconnect layer can be coupled to the contact structure. The interconnect layer can include at least an interconnect line (e.g., the interconnect lines **246** of FIG. **2D**) extending along a second direction perpendicular to the first direction and a VIA (e.g., one of the VIAs **242** or **248** of FIG. **2D**) extending along the first direction.

[0079] At operation **306**, the first semiconductor structure is bonded to a second semiconductor structure (e.g., the semiconductor structure **258** of FIG. **2E**). The second semiconductor structure can include a peripheral circuit (e.g., the peripheral circuit **268** of FIG. **2E**) associated with the array of memory cells. The contact structure is coupled to the peripheral circuit. In some implementations, the carrier wafer can be removed or de-bonded from the first semiconductor structure. In some implementations, the first semiconductor structure can be bonded to the second semiconductor structure through a bonding structure (e.g., the bonding structure **266** of FIG. **2E**).

The bonding structure includes conductive bonding pads (e.g., the conductive bonding pads **252** and **262**) and at least one dielectric material (the dielectric materials **254** and **264**) isolating the conductive bonding pads in a second direction perpendicular to the first direction. In some implementations, the first side (e.g., the side **245** of FIG. 2E) of the first semiconductor structure is between the second side (e.g., the side **230** of FIG. 2E) of the first semiconductor structure and the bonding structure.

[0080] In some implementations, the process **300** further includes forming a trench (e.g., the trench **274** of FIG. 2F) extending from the second side of the first semiconductor structure into the first semiconductor structure. In some implementations, the process **300** further includes forming a conductive contact structure (e.g., the contact structure **278** of FIG. 2G) in the trench. For example, the conductive contact structure can be formed by depositing a conductive material (e.g., Al) in the trench. The conductive contact structure can be coupled to the storage structure.

[0081] FIG. 4 illustrates a block diagram of an example system **400**. The system **400** can have one or more semiconductor devices (e.g., memory devices), according to one or more implementations of the present disclosure. The system **400** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. 4, the system **400** can include a host device **408** and a memory system **402** having one or more memory devices **404** and a memory controller **406**. Host device **408** can include a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host device **408** can be configured to send or receive data to or from the one or more memory devices **404**.

[0082] A memory device **404** can be any memory device disclosed herein, such as a memory device (e.g., a DRAM device) as shown in FIGS. 1 and 2A-2G. In some implementations, a memory device **404** includes a NAND Flash memory. Memory controller **406** (a.k.a., a controller circuit) is coupled to memory device **404** and host device **408**. Consistent with implementations of the present disclosure, memory device **404** can include a plurality of conductive interconnections through a cover layer that are in contact with conductive pads in a conductive pad layer, and memory controller **406** can be coupled to memory device **404** through at least one of the plurality of conductive interconnections. Memory controller **406** is configured to control memory device **404**. For example, memory controller **406** may be configured to operate a plurality of channel structures via word lines. Memory controller **406** can manage data stored in memory device **404** and communicate with host device **408**.

[0083] In some implementations, memory controller **406** is designed/configured for operating in a low duty-cycle environment like secure digital (SD) cards, compact Flash (CF) cards, universal serial bus (USB) Flash drives, or other media for use in electronic devices, such as personal computers, digital cameras, mobile phones, etc. In some implementations, memory controller **406** is designed/configured for operating in a high duty cycle environment SSDs or embedded multi-media-cards (eMMCs) used as data storage for mobile devices, such as smartphones, tablets, laptop computers, etc., and enterprise storage arrays. Memory controller **406** can be configured to control operations of memory device **404**, such as read, erase, and program (or write) operations. Memory controller **406** can also be configured to manage various functions with respect to the data stored or to be stored in memory device **404** including, but not limited to bad-block management, garbage collection, logical-to-physical address conversion, wear leveling, etc. In some implementations, memory controller **406** is further configured to process error correction codes (ECCs) with respect to the data read from or written to memory device **404**. Any other suitable functions may be performed by memory controller **406** as well, for example, formatting memory device **404**.

[0084] Memory controller **406** can communicate with an external device (e.g., host device **408**) according to a particular communication protocol. For example, memory controller **406** may

communicate with the external device through at least one of various interface protocols, such as a USB protocol, an MMC protocol, a peripheral component interconnection (PCI) protocol, a PCI express (PCI-E) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

[0085] Memory controller **406** and one or more memory devices **404** can be integrated into various types of storage devices, for example, be included in the same package, such as a universal Flash storage (UFS) package or an eMMC package. That is, memory system **402** can be implemented and packaged into different types of end electronic products. In one example as shown in FIG. 4, memory controller **406** and a single memory device **404** may be integrated into a memory card **402**. Memory card **402** can include a PC card (PCMCIA, personal computer memory card international association), a CF card, a smart media (SM) card, a memory stick, a multimedia card (MMC, RS-MMC, MMCmicro), an SD card (SD, miniSD, microSD, SDHC), a UFS, etc.

[0086] Implementations of the subject matter and the actions and operations described in this present disclosure can be implemented in digital electronic circuitry, in tangibly-embodied computer software or firmware, in computer hardware, including the structures disclosed in this present disclosure and their structural equivalents, or in combinations of one or more of them. Implementations of the subject matter described in this present disclosure can be implemented as one or more computer programs, e.g., one or more modules of computer program instructions, encoded on a computer program carrier, for execution by, or to control the operation of, data processing apparatus. The carrier may be a tangible non-transitory computer storage medium. Alternatively, or in addition, the carrier may be an artificially-generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal, that is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus. The computer storage medium can be or be part of a machine-readable storage device, a machine-readable storage substrate, a random or serial access memory device, or a combination of one or more of them. A computer storage medium is not a propagated signal.

[0087] It is noted that references in the present disclosure to “one embodiment,” “an embodiment,” “an example embodiment,” “some implementations,” “some implementations,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of a person skilled in the pertinent art to affect such feature, structure or characteristic in connection with other implementations whether or not explicitly described.

[0088] In general, terminology can be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, can be used to describe any feature, structure, or characteristic in a singular sense or can be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, can be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” can be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

[0089] It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something, but also includes the meaning of “on” something with an intermediate feature or a layer therebetween. Moreover, “above” or “over” not only means “above” or “over” something, but can

also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (i.e., directly on something).

[0090] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, can be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or process step in addition to the orientation depicted in the figures. The apparatus can be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein can likewise be interpreted accordingly.

[0091] As used herein, the term “substrate” refers to a material onto which subsequent material layers are added. The substrate includes a “top” surface and a “bottom” surface. The top surface of the substrate is typically where a semiconductor device is formed, and therefore the semiconductor device is formed at a top side of the substrate unless stated otherwise. The bottom surface is opposite to the top surface and therefore a bottom side of the substrate is opposite to the top side of the substrate. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

[0092] As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer has a top side and a bottom side where the bottom side of the layer is relatively close to the substrate and the top side is relatively away from the substrate. A layer can extend over the entirety of an underlying or overlying structure, or can have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any set of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layer thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductive and contact layers (in which contacts, interconnect lines, and/or vertical interconnect accesses (VIAs) are formed) and one or more dielectric layers.

[0093] As used herein, the term “nominal/nominally” refers to a desired, or target, value of a characteristic or parameter for a component or a process step, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. As used herein, the range of values can be due to slight variations in manufacturing processes or tolerances. As used herein, the term “about” indicates the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. Based on the particular technology node, the term “about” can indicate a value of a given quantity that varies within, for example, 10-30% of the value (e.g., $\pm 10\%$, $\pm 20\%$, or $\pm 30\%$ of the value).

[0094] In the present disclosure, the term “horizontal/horizontally/lateral/laterally” means nominally parallel to a lateral surface of a substrate, and the term “vertical” or “vertically” means nominally perpendicular to the lateral surface of a substrate.

[0095] As used herein, the term “3D memory” refers to a three-dimensional (3D) semiconductor device with vertically oriented strings of memory cell transistors (referred to herein as “memory strings,” such as NAND strings) on a laterally-oriented substrate so that the memory strings extend in the vertical direction with respect to the substrate.

[0096] The present disclosure provides many different implementations, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course,

merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include implementations in which the first and second features may be in direct contact, and may also include implementations in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various implementations and/or configurations discussed.

[0097] The foregoing description of the specific implementations can be readily modified and/or adapted for various applications. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed implementations, based on the teaching and guidance presented herein.

[0098] While the present disclosure contains many specific implementation details, these should not be construed as limitations on the scope of what is being claimed, which is defined by the claims themselves, but rather as descriptions of features that may be specific to particular implementations of particular inventions. Certain features that are described in this present disclosure in the context of separate implementations can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially be claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claim may be directed to a sub-combination or variation of a sub-combination.

[0099] Similarly, while operations are depicted in the drawings and recited in the claims in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system modules and components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

[0100] Particular implementations of the subject matter have been described. Other implementations also are within the scope of the following claims. For example, the actions recited in the claims can be performed in a different order and still achieve desirable results. As one example, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In some cases, multitasking and parallel processing may be advantageous.

[0101] The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary implementations, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A semiconductor device, comprising a first semiconductor structure and a second semiconductor structure connected together, wherein: the first semiconductor structure comprises a first contact structure extending along a first direction and an array of memory cells; the first contact structure comprises a first cross section and a second cross section both perpendicular to the first direction; the first cross section is farther away from the second semiconductor structure than the second cross section along the first direction; and a size of the first cross section is smaller than a size of

the second cross section.

2. The semiconductor device of claim 1, wherein the second semiconductor structure comprises a peripheral circuit associated with the array of memory cells, and the peripheral circuit is coupled to the first contact structure.

3. The semiconductor device of claim 1, wherein the size of the first cross section is a maximum size of the first cross section along a second direction perpendicular to the first direction, and the size of the second cross section is a maximum size of the second cross section along the second direction.

4. The semiconductor device of claim 1, wherein the first contact structure extends through a semiconductor layer of the first semiconductor structure, and the first contact structure is spaced from the array of memory cells along a second direction perpendicular to the first direction.

5. The semiconductor device of claim 1, further comprising a bonding structure between the first semiconductor structure and the second semiconductor structure, wherein: the second semiconductor structure is bonded to a first side of the first semiconductor structure through the bonding structure; and the bonding structure comprises conductive bonding pads and at least one dielectric material isolating the conductive bonding pads in a second direction perpendicular to the first direction.

6. The semiconductor device of claim 5, wherein: the first semiconductor structure further comprises an interconnect layer between the first contact structure and the bonding structure along the first direction; a first end of the first contact structure is coupled to one of the conductive bonding pads of the bonding structure through the interconnect layer; and a second end of the first contact structure is coupled to a first conductive contact structure on a second side of the first semiconductor structure, the second side being opposite to the first side.

7. The semiconductor device of claim 6, wherein: the interconnect layer comprises at least an interconnect line extending along the second direction and a vertical interconnect access (VIA) extending along the first direction; the VIA comprises a third cross section and a fourth cross section both perpendicular to the first direction; the third cross section is farther away from the second semiconductor structure than the fourth cross section along the first direction; and a size of the third cross section is smaller than a size of the fourth cross section.

8. The semiconductor device of claim 6, wherein: at least one of the array of memory cells comprises a transistor extending along the first direction and a storage structure; a first end of the storage structure is coupled to the transistor; a second end of the storage structure is coupled to a second conductive contact structure in contact with a trench, the trench extending from the second side of the first semiconductor structure into the first semiconductor structure; and both the first conductive contact structure and the second conductive contact structure comprise aluminum.

9. The semiconductor device of claim 8, wherein: the second conductive contact structure comprises a first portion, a second portion, and a third portion; the first portion and the second portion are connected by the third portion; the first portion and the second portion are perpendicular to the first direction; the first portion is in contact with a bottom of the trench; the third portion is in contact with sidewalls of the trench; and the second portion is in contact with a part of the second side that is connected to the sidewalls of the trench.

10. The semiconductor device of claim 8, wherein a ratio between a first size of the trench in the first direction and a second size of the trench in the second direction is smaller than 0.5.

11. The semiconductor device of claim 2, wherein: the first semiconductor structure further comprises a second contact structure extending along the first direction through a semiconductor layer of the first semiconductor structure; the second contact structure is coupled to the peripheral circuit; the first contact structure is configured to transfer a control signal to and from the peripheral circuit; and the second contact structure is configured to provide power to the peripheral circuit.

12. A method, comprising: forming a first semiconductor structure comprising an array of memory

cells, wherein at least one of the array of memory cells comprises a transistor extending along a first direction and a storage structure coupled to the transistor; forming a contact structure in the first semiconductor structure, wherein the contact structure extends along the first direction; and bonding the first semiconductor structure to a second semiconductor structure comprising a peripheral circuit associated with the array of memory cells, wherein the contact structure is coupled to the peripheral circuit.

13. The method of claim 12, wherein forming the first semiconductor structure comprising the array of memory cells comprises: forming a semiconductor body of the transistor in a semiconductor layer of the first semiconductor structure; and forming a dielectric region extending through the semiconductor layer along the first direction.

14. The method of claim 13, wherein the first semiconductor structure comprises a first side and a second side opposite to the first side, the first side is closer to the semiconductor layer than the second side along the first direction, and forming the contact structure in the first semiconductor structure comprises: forming a contact hole in the first semiconductor structure by etching the first semiconductor structure from the first side, wherein the contact hole extends through the dielectric region; and forming the contact structure in the contact hole by depositing a conductive material into the contact hole.

15. The method of claim 14, further comprising: stacking the first semiconductor structure on a carrier wafer before forming the contact hole, wherein the second side is in contact with the carrier wafer.

16. The method of claim 14, further comprising: forming an interconnect layer in the first semiconductor structure, wherein the interconnect layer is coupled to the contact structure and comprises at least an interconnect line extending along a second direction perpendicular to the first direction and a vertical interconnect access (VIA) extending along the first direction.

17. The method of claim 14, wherein bonding the first semiconductor structure to the second semiconductor structure comprising: bonding the first semiconductor structure to the second semiconductor structure through a bonding structure, wherein the bonding structure comprises conductive bonding pads and at least one dielectric material isolating the conductive bonding pads in a second direction perpendicular to the first direction, and the first side is between the second side and the bonding structure.

18. The method of claim 14, further comprising: forming a trench extending from the second side of the first semiconductor structure into the first semiconductor structure; forming a conductive contact structure in the trench by depositing a conductive material in the trench, wherein the conductive contact structure is coupled to the storage structure.

19. A memory system, comprising: a memory device; and a memory controller coupled to the memory device and configured to control the memory device, wherein: the memory device comprising a first semiconductor structure and a second semiconductor structure connected together; the first semiconductor structure comprises a first contact structure extending along the first direction and an array of memory cells; the first contact structure comprises a first cross section and a second cross section both perpendicular to the first direction; the first cross section is farther away from the second semiconductor structure than the second cross section along the first direction; and a size of the first cross section is smaller than a size of the second cross section.

20. The memory system of claim 19, wherein the size of the first cross section is a maximum size of the first cross section along a second direction perpendicular to the first direction, and the size of the second cross section is a maximum size of the second cross section along the second direction.
